A-A-1646
July 26, 1982

COMMERCIAL ITEM DESCRIPTION
Household Ranges, Gas (Domestic)

The General Services Administration has authorized the use of this Commercial Item Description in preference to Federal Specification S-R-440

This commercial item description covers the requirements for household gas ranges utilizing automatic ignition.

Salient Characteristics

Household gas ranges covered by this CID shall be of the following sizes and types and meet the minimum thermal efficiencies.

Size	Type	Surface Burners	Oven
20 - 20½ ± 1 inch wide	Single Oven	43%	6%
30 - 20 ± 1 inch wide	Single Oven	43%	5.4%
36 - 36 ± 1 inch wide	Single Oven	43%	5.4%
40 - 40 ± 1 inch wide	Double Oven	43%	-
	Small Oven		6%
	Large Oven		5.4%

The surface burner thermal efficiency is based on the average of the four surface burner thermal efficiencies.

The efficiencies shall be measured in accordance with the Department Energy test method for Conventional Ranges, Cooking Tops, Ovens, Microwave Ovens, Microwave/Conventional Ranges as written in the Federal Register, Vol. 43, No. 91 - Wednesday, May 10, 1978, pages 20108 thru 20128.

The Ranges shall be free-standing, designed for flush-to-wall installation, and for operation with natural gas and liquidfied petroleum gas.

Lights, clocks, and other electrical equipment on the ranges shall operate on a normal 120 volt, 60 hz, single phase alternating current system.

FSC 7310

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The oven door hinges shall be able to pass an oven door hinge test. This test is as follows:

The oven thermostat shall be set at 330 ± 5°F above ambient temperature and oven operated for at least 2 hours to assure stabilization of the oven temperature. Then without adjustment of the oven controls, the oven door shall be operated through 25,000 cycles from an essentially closed position to an essentially fully open position and returned at a frequency of approximately 15 cycles per minute. Upon completion of the 25,000 cycles, the oven shall be shut down and, without adjustment, modification or disassembly of the oven door, its hinges, or any other component part, the oven door shall be subject to a load of 25 pounds (± 5 percent) applied without impact to a 5 square inch (± 10 percent) area for 5 minutes at one outer corner of the fully open door while the other outer corner is fully supported. The load shall then be removed, the door examined for proper fit and closure at which time, there shall be no warpage of the oven door or evidence of gaps between the oven face the oven door when fully closed position. A distributed load of 50 pounds (± 5 percent) shall then be applied without impact along a 4 inch (± 10 percent) strip midway between the hinged edge and the outer edge of the fully open door for 5 minutes. At the conclusion of the 5 minute period, without adjust to the door or its components, the door shall be closed and the surface temperature test described in the latest edition of ANSI Z21.1 shall be conducted.

Following the Surface Temperature Test, the oven door and its components shall be examined and shall exhibit no evidence of breakage or permanent distortion. Any nonconformance of these requirements shall constitute a failure of the test.

Except where otherwise specified, all ranges shall be provided with the following features:

a) Porcelain enameled oven lining with two corrosion-resisting steel, or nickel-plated or chrome-nickel-plated racks;

b) Broiler compartment, broiler pan and rack and base baffle finishes shall be in accordance with the current industry standard practice.

c) Recessed (no drip type) cooking top of the lift-off or lift-up type, all size ranges shall have a porcelain finish on the burner box or equivalent.

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d) Clock and an interval timer which sounds an alarm when the pre-set time has elapsed.

e) On the double oven range, one (1) oven shall be controlled by an automatic clock oven device. This device, shall be the manufacturers commercially current standard feature.

f) A manual switch controlled light in all ovens, on double oven ranges, one manual switch may control both oven lights.

g) A heat-resisting, oven viewing window in all oven doors except in the oven door of the size 20 range;

h) Four top burners with a minimum burner rating of 9000 BTU/hr for natural gas and 8000 BTU/hr for liquified petroleum gas;

i) A decorative backguard except for size 20 ranges;

j) At least two leveling devices capable of leveling the ranges when installed.

The gas ranges shall conform to ANSI Z 21.1. The automatic gas ignition system for the ranges shall meet the applicable requirements of ANSI Z 21.20 Acceptable evidence of meeting the applicable requirements of these standards shll be a photostatic copy of the AGA Approval Certificate, a listing of the igniter device in the AGA Director of Certified Applicances and Accessories, or a certified test report from a recognized, independent testing laboratory acceptable to the Government stating that the range conforms to ANSI Z 21.20.

Porcelain enamel finished cooktops shall be examined and shall have a Class A or better acid resistance when tested in accordance with ASTM method C 282. Porcelain enamel thickness of cooktops, door and other front panels shall not be greater than 12 mils.

All ovens shall have at least 10 inches of "usuable oven space" measured vertically from the highest projection of an oven rack in its lowest position to the lowest projection of the oven ceiling. (Excluding upper oven on double-oven ranges).

Surfaces of parts not exposed to view in normal usage shall be protected in such a manner that any oxidation which may occur will not interfere with its intended function. An oxide coating is acceptable for these exposed surfaces.

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Workmanship.
All ranges shall be designed in such a manner as to permit ease of cleaning all surfaces and cavities where grease and other forms of soil collect. The ranges shall be so designed that parts mate properly without excessive play. The finished ranges shall not contain any burrs, slivers, sharp edges, or rough dye, tool gouge, or grind marks. Holes for bolts, screws, rivets, and threaded fasteners shall be accurately punched or drilled, and any obstructions that prevent proper seating of rivets shall be removed. Driven rivets shall be fully peened with the heads in full contact with riveted members. Components shall not be fractured, buckled, kinked, dented, punctured, or malformed. There shall be no evidence of pits or stains, blistering, flaking, peeling, or large discoloration (exposed surfaces). Components shall be free of sags, runs, areas of no film, rust, imbedded foreign matter, and wet or tacky painted surfaces. (Exposed surfaces)

Recovered Materials.
The manufacturer shall utilize recovered materials to the maximum extent practicable.

Packaging, Packing, and Marking.
The household ranges shall be packaged in accordance with normal commercial practice and packed in fiberboard boxes.

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